



Evaluating the Accuracy and Stability of Frontier LLMs on ENADE Computer Science Questions

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Abstract. Large Language Models (LLMs) are increasingly evaluated on complex reasoning tasks, yet most benchmarks rely on single-shot prompting and fail to account for output variability. This study investigates both the accuracy and response stability of four multimodal frontier LLMs—GPT-4o, o1, Gemini 2.0 Flash, and DeepSeek-R1—on the multiple-choice Computer Science section of Brazil’s 2021 ENADE exam. Each model was prompted ten independent times per question, and their performance was evaluated using five complementary accuracy and stability metrics. Our results reveal substantial variation in both accuracy and stability across models and questions. Gemini 2.0 Flash achieved the highest accuracy, while also exhibiting strong response stability, followed by o1 and DeepSeek-R1. In contrast, GPT-4o showed lower accuracy and notably less stability. Interestingly, higher stability did not always correlate with higher accuracy, with some models being more stable than others, but less accurate. A per-question analysis shows that questions excluded from the ENADE score due to psychometric anomalies often correlate with high model disagreement or systematic errors, though some excluded questions were answered consistently by most models, providing insights into LLMs as complementary tools for exam validation. These findings reinforce the importance of multi-run evaluations and stability-focused metrics when benchmarking LLMs, especially in high-stakes educational contexts.

Keywords: Large language models · Accuracy · Stability · ENADE

1 Introduction

The emergence of large language models (LLMs), such as OpenAI’s ChatGPT and Google’s Gemini, has led to a paradigm shift in natural language processing. These models demonstrate remarkable performance in tasks requiring abstract reasoning, problem-solving, programming, and domain-specific knowledge [29].

Their subsequent integration of multimodal capabilities [22] extends their potential to educational contexts where visual content, such as diagrams, charts, and pseudocode, plays a central role in assessment and learning [8].

In academic settings, LLMs present both opportunities and challenges [14, 15]. For students, they offer new tools for personalized learning and exam preparation [7]; for educators, they serve as a resource for generating or refining instructional and assessment material [30]; and for exam designers, they can aid in identifying ambiguous or low-quality questions before test deployment [18]. However, LLMs’ non-deterministic nature and occasional reasoning failures also pose risks, especially in high-stakes educational contexts [19].

A recent study [18] evaluated GPT-4 Vision on Brazil’s 2021 ENADE Computer Science (BCS) exam, a national assessment for undergraduate programs. While the model showed strong single-run performance, the evaluation was limited to one model and attempt per question, preventing an analysis of response stability, an increasingly important consideration for trustworthy AI [2]. This study extends that work by evaluating four frontier multimodal LLMs (GPT-4o, o1, Gemini 2.0 Flash, and DeepSeek-R1) on the same ENADE questions **over ten independent runs per model per question**. This multi-run approach allows us to investigate both accuracy and stability, quantifying how often models “get it right eventually” versus “get it right consistently,” a critical distinction in high-stakes educational contexts.

To this end, we employ five complementary evaluation metrics: *mean accuracy*, which computes the proportion of correct responses across all attempts; *pass@N*, which checks whether the correct answer appeared at least once; *accuracy spread*, which measures the difference between the maximum and minimum mean accuracy across attempts; *Total Agreement Rate (TAR@N)*, which checks whether all responses were identical (regardless of correctness); and *consensus correctness (cons@N)*, which checks whether the unique most frequent answer was correct. These metrics—drawn from prior work in model evaluation [2, 4] and industry practice [23] and applied here for exam assessment—help uncover both systematic errors and inconsistent behavior in current frontier LLMs, providing a more complete view of model trustworthiness in educational contexts.

Our results reveal that Gemini 2.0 Flash consistently achieves the highest accuracy and stability across all five metrics, followed by o1 and DeepSeek-R1. GPT-4o exhibits both the lowest accuracy and highest variability, occasionally producing correct answers but failing to do so consistently. Interestingly, higher consistency did not always correlate with higher accuracy: DeepSeek-R1 was more stable than o1, but less accurate. A per-question analysis shows that questions excluded from the ENADE score due to psychometric anomalies often correlate with high model disagreement or systematic errors, though some excluded questions were answered reliably by most models, revealing divergence between psychometric and LLM-based quality testing [5].

In summary, this paper makes the following contributions:

- We conduct a comprehensive multi-model, multi-run evaluation of LLM performance on a high-stakes, multimodal educational benchmark (ENADE 2021 BCS exam).
- We apply five complementary accuracy and stability metrics to assess both model correctness and response consistency.
- We analyze model per-question behaviors to identify systematic failures, misleading distractors, and potential question design issues.
- We examine how model performance correlates with questions excluded by exam organizers via psychometric analysis, providing insights into LLMs as complementary tools for exam validation.

The remainder of this paper is structured as follows. Section 2 reviews relevant related work. Section 3 details our research methodology. Section 4 presents our findings. Section 5 discusses the broader implications and practical limitations of our work. Finally, Sect. 6 offers our conclusions and future research directions.

2 Related Work

2.1 LLM Evaluation in Educational Domains

Recent studies have evaluated the reasoning capabilities of LLMs on academic benchmarks across educational domains such as medicine [20], engineering [9], and computer science [3, 27]. While GPT-4 frequently outperforms prior models on multiple-choice exams, it still exhibits notable limitations in tasks requiring symbolic reasoning, code synthesis, or multi-step logic [16]. Within computer science education specifically, ChatGPT-3.5 and GPT-4 have been tested on curriculum-aligned assessments and software engineering problems [12, 13], with mixed results depending on task complexity and format. These works underscore the value of domain-specific benchmarks for probing LLM capabilities under realistic academic conditions.

2.2 Stability and Non-determinism in LLM Responses

A growing body of research investigates the instability of LLM outputs under multiple generations. For example, Atil *et al.* found that no state-of-the-art LLM achieved fully repeatable accuracy across multiple tasks, with up to 10% variation across identical runs [2]. Ouyang *et al.* observed that in most code-generation benchmarks, ChatGPT’s outputs differed across attempts in over half the problems, even with model temperature set to zero [24]. These findings reinforce the need for multi-run evaluations and metrics that explicitly capture internal consistency. Our study incorporates this perspective by using metrics such as accuracy spread, TAR@N, and cons@N to quantify variability, intra-question determinism, and consensus-based correctness across repeated trials.

2.3 Brazil’s Educational Assessment Benchmarks

Recent studies have begun to assess the performance of LLMs on Brazil’s national exams, including ENEM (a multidisciplinary university entrance exam), ENADE, and POSCOMP (a graduate-level Computer Science entrance exam). GPT-4 has shown strong performance on ENEM [25] and ENADE [21] multiple-choice questions, particularly when aided by human-generated captions for inputs containing visual elements. Saldanha and Digiampietri [28] evaluated ChatGPT and Bard on hundreds of POSCOMP questions and found that both models exceeded average human performance, even without additional prompt context. Mendonça [18] conducted a single-shot evaluation of GPT-4 Vision on the 2021 ENADE Computer Science exam, using the original visual format, and reported high alignment with the official answer key.

These studies highlight the growing interest in benchmarking LLMs using national educational assessments but focus exclusively on single-shot accuracy. Our work builds on this foundation by comparing four frontier models on the same ENADE exam evaluated by Mendonça and introducing response stability as a key evaluation dimension, using ten independent runs per question to reveal intra-model variability.

3 Methodology

This section describes the experimental design adopted in this study, including the evaluation benchmark, research questions, models investigated, experimental protocol, and performance metrics.

3.1 Evaluation Benchmark

The benchmark used in this study is the Bachelor in Computer Science (BCS) section of Brazil’s 2021 National Undergraduate Exam (*Exame Nacional de Desempenho dos Estudantes—ENADE*) [10].¹ The 2021 BCS exam comprises 30 open-ended (3) and multiple-choice (27) questions covering various Computer Science topics. ENADE organizers annulled two multiple-choice question after the exam had been administered. This study focuses on the 25 remaining valid multiple-choice questions, which often include visual elements like diagrams, charts, code snippets, and algorithms alongside textual descriptions. Six of the valid questions ended up being excluded from the official ENADE scoring due to potential quality issues identified via psychometric analysis of the participant’s performance [11]. However, these excluded questions were included in our analysis to assess the models’ performance on these potentially problematic items.

¹ The 2021 edition was the last time the ENADE BCS exam was administered to Brazilian undergraduate students at the time of writing this paper.

3.2 Research Questions

The goal of this study is to examine how accurately and consistently modern frontier LLMs perform on the 2021 BCS ENADE exam under multiple attempts per question. We pursue this goal through the following four research questions:

RQ1 (Accuracy): How accurate are LLMs when evaluated over multiple runs?

RQ2 (Stability): How consistent are LLM responses under repeated sampling?

RQ3 (Per-Question Performance): How does the models' performance vary across individual questions?

RQ4 (Excluded Questions): Do model per-question performance patterns align with questions excluded from the official ENADE scoring due to statistical anomalies?

3.3 Selected Models and Experimental Protocol

We evaluated four multimodal large language models (LLMs) considered to be at the technological frontier at the time of the study (late 2024 to early 2025): OpenAI's GPT-4o and o1, Google's Gemini 2.0 Flash, and DeepSeek's DeepSeek-R1, the latter being the only open-weight model of the batch. These models were selected based on their advanced reasoning capabilities, particularly in handling multimodal inputs (relevant to the visual nature of some ENADE questions) and their prominence in the field. According to industry insiders,² these four models have cutoff dates for training data from May 13, 2024 (GPT-4o) to January 20, 2025 (DeepSeek-R1). This means that the ENADE 2021 exam and its official answer key could have been publicly available during the models' training time, which might have influenced the models' performance. We discuss this and other validity threats in Sect. 5.2.

While several open-weight or fine-tuned LLMs tailored to the Portuguese language or Brazilian educational context have recently emerged, such as Sabiá-3 [1] and Amadeus-Verbo [6], our study focused on evaluating frontier-level multimodal models that represent the state of the art in general-purpose reasoning and visual understanding. To the best of our knowledge, there are currently no top-tier multimodal models with specific training or fine-tuning for Brazilian Portuguese capable of processing images. Nevertheless, evaluating such models, once available, would provide a valuable complementary perspective and is an important direction for future work.

We presented each of the 25 valid multiple-choice questions of the 2021 BCS ENADE exam to each model in image format, including any text and visual elements. We created each image by taking a high-resolution screen shot of the question directly from the exam's PDF file [10]. To assess response stability and mitigate the effects of inherent non-determinism, we prompted each model to answer each question 10 independent times. For each attempt, we asked the model to select one of the five options in the question (A-E)—or to respond

² <https://explodingtopics.com/blog/list-of-llms>.

Table 1. Accuracy and stability metrics used in our study.

Category	Metric	Operational Definition
Accuracy	Mean Accuracy	Computes the rate of correct answers across N responses for a question. The overall mean accuracy is the average over all questions.
	pass@N	Binary metric that returns 1 if there is at least one correct answer across N responses for a question or 0 otherwise. The overall pass@N is the average over all questions. Higher values indicate the model is more likely to produce a correct response eventually.
Stability	Accuracy Spread	Computes the difference between the maximum and minimum mean accuracy across N attempts over all questions. Lower values indicate the model is more accurately stable across attempts.
	TAR@N	Binary metric that returns 1 if all N answers for a question are identical, regardless of their correctness, or 0 otherwise. The overall TAR@N is the average over all questions. Higher values indicate the model produces more self-consistent responses across attempts.
	cons@N	Binary metric that returns 1 if the unique most frequent answer (the <i>mode</i> or “consensus”) across N responses for a question is correct or 0 otherwise. The overall cons@N is the average over all questions. Higher values indicate the model is more likely to produce a correct response consistently.

with option “X” if the model concluded that the question had more than one correct answer or no correct answer—and provide a step-by-step explanation for its reasoning. We compared each response generated by a model against the official ENADE answer key to ascertain whether the model was correct. We interacted with all models via their web-based interfaces, using the default settings for the temperature and other configuration parameters. This decision was made to ensure that the models were evaluated in their most natural and user-friendly state, as they would likely be used by students and educators in real-world scenarios.

3.4 Performance Metrics

To evaluate both accuracy and stability of the models’ responses across multiple runs, we computed the following performance metrics: *mean accuracy*, *accuracy spread*, *pass@N*, *Total Agreement Rate (TAR@N)*, and *consensus correctness (cons@N)*. Table 1 provides the operational definition for each of these five metrics. The mean accuracy, accuracy spread, and TAR@N metrics were used to measure model accuracy and stability in a recent work by Atil *et al.* [2]. pass@N

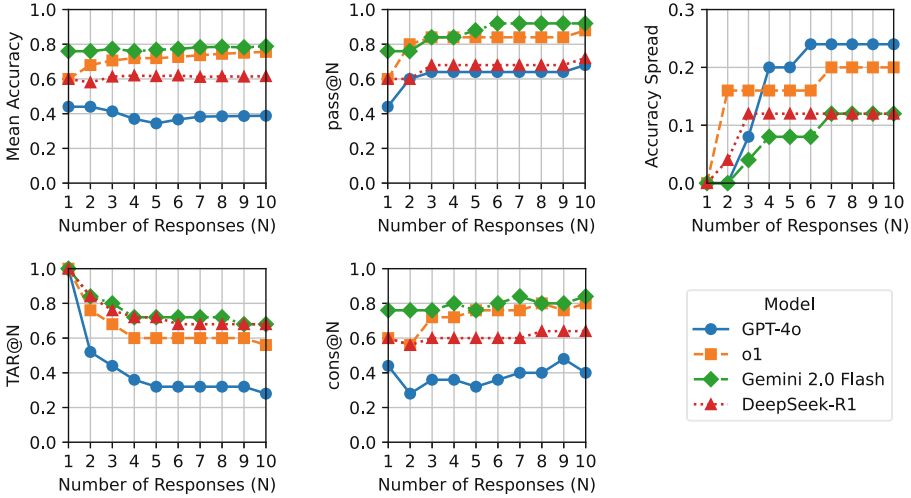


Fig. 1. Average performance of the four models across 10 responses over all questions for five metrics: mean accuracy (top-left), pass@N (top-center), accuracy spread (top-right), TAR@N (bottom-left), and cons@N (bottom-center).

and cons@N are standard measures of model performance commonly used in industry-led benchmarks (e.g., [4, 23]).

4 Results

We first analyze how the accuracy and stability of the four models investigated evolve as the number of responses per question increases from 1 to 10. Figure 1 shows five plots representing the accumulative average performance of the four models over all 25 multiple-choice questions for each of the five metrics described in Table 1. These results will be used to address the first two research questions (RQ1 and RQ2) by providing a comprehensive overview of the models’ accuracy and stability across multiple response generations.

4.1 RQ1: Accuracy

Regarding mean accuracy (Fig. 1, top-left), Gemini 2.0 Flash consistently leads with the highest values, approaching 0.8 as the number of responses grows. It is closely followed by o1, which gradually converges to ≈ 0.75 . DeepSeek-R1 maintains a slightly lower (≈ 0.6) but steady average performance, while GPT-4o underperforms compared to the other models, showing a mild decrease in accuracy as N increases and finally stabilizing at about 0.4. This suggests that repeated sampling does not substantially improve GPT-4o’s accuracy and may even introduce occasional regressions.

In terms of pass@N (Fig. 1, top-center), which evaluates whether at least one of the multiple responses for a question is correct, all models benefit from additional attempts, but their performance gains vary. Gemini 2.0 Flash and o1 reach average pass@N values close to 0.9, indicating high likelihood of generating at least one correct answer within ten attempts per question. DeepSeek-R1 levels off at around 0.72, closely followed by GPT-4o, which peaks slightly below 0.7. These trends reinforce the models' ranking observed in mean accuracy, but with a more forgiving accuracy metric.

Overall, Gemini 2.0 Flash is the most accurate model, achieving the highest mean accuracy and pass@N rates, followed closely by o1. DeepSeek-R1 performs moderately well in both metrics, while GPT-4o lags behind the others, particularly in the mean accuracy metric.

4.2 RQ2: Stability

For accuracy spread (Fig. 1, top-right), which measures the variability in accuracy across multiple attempts, GPT-4o and o1 display the highest values, reaching about 0.24 and 0.2, respectively, after the 10th response, suggesting moderate instability in their correct answers. In contrast, Gemini 2.0 Flash and DeepSeek-R1 show 50% lower accuracy spread throughout (≈ 0.12) compared to the other models, demonstrating much better consistency. These results indicate that Gemini 2.0 Flash is not only the most accurate but also one of the two most stable of the four models across multiple attempts. Curiously, the more stable accuracy results for DeepSeek-R1 do not necessarily translate into higher mean accuracy overall, indicating that accuracy and stability are not always directly related.

TAR@N (Fig. 1, bottom-left), which measures the percentage of questions where all answers are identical, drops quickly for all models as the number of responses increases. Gemini 2.0 Flash and DeepSeek-R1 show the highest TAR@N values, stabilizing around 0.7, indicating that they tend to produce identical answers across multiple attempts. o1 follows closely behind, with a TAR@N of about 0.6, suggesting that it also generates consistent responses but with slightly more variability. GTP-4o, however, shows a steep decline in TAR@N, dropping to about 0.3 after the 5th response and below that after the last one. This indicates that GPT-4o generates more diverse responses across attempts, leading to lower consistency in its answers.

Finally, under cons@N (Fig. 1, bottom-center), which evaluates whether the most frequent answer is correct, Gemini 2.0 Flash again leads with average values consistently at or above 0.8, followed closely by o1. DeepSeek-R1 shows moderate consensus (leveraging out at ≈ 0.6), while GPT-4o has the lowest values (around 0.3–0.5), reflecting its higher output variability. This suggests that GPT-4o may occasionally produce correct answers, but it does so with less consistency and agreement across attempts.

Overall, Gemini 2.0 Flash and DeepSeek-R1 are the most stable models, with low accuracy spread and high TAR@N. In contrast, GPT-4o shows the highest variability in its responses, leading to lower stability and consensus. The

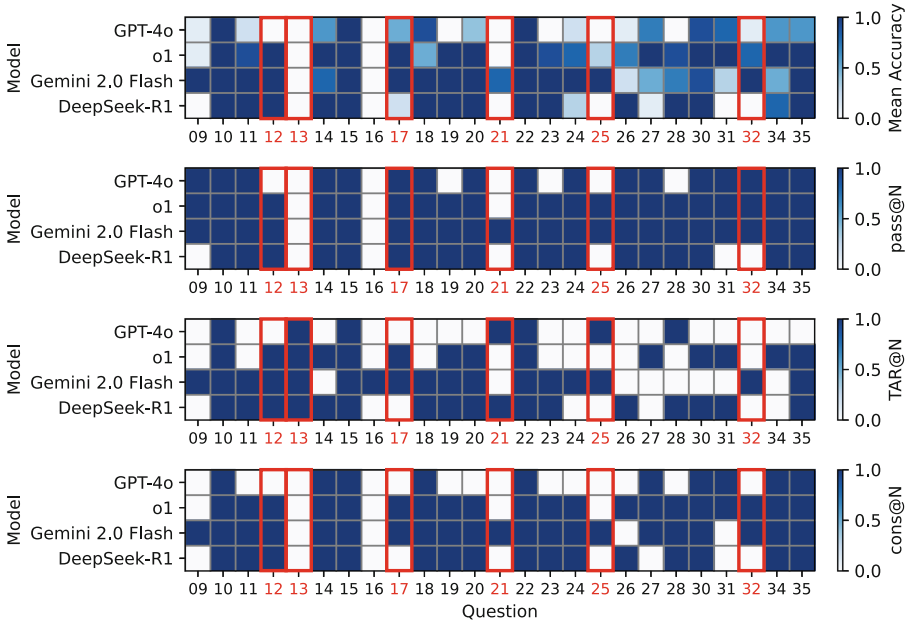


Fig. 2. Per-question performance of the four models over 10 responses for four metrics (from top to bottom): mean accuracy, pass@N, TAR@N, and cons@N. Questions excluded from the official exam score are highlighted in red. (Color figure online)

performance of o1 falls between these two extremes, demonstrating moderate stability and accuracy.

In summary, considering all five metrics, Gemini 2.0 Flash emerges as the most accurate and stable model in this benchmark. It consistently achieves high mean accuracy, pass@N, and TAR@N across multiple attempts, while also maintaining a low accuracy spread. o1 also performs well, particularly in terms of accuracy, but lags slightly behind in stability. DeepSeek-R1 shows good stability by only moderate performance in accuracy. Finally, GPT-4o exhibits the weakest performance amongst the four models investigated, displaying moderate to low results in both accuracy and stability.

4.3 RQ3: Per-Question Performance

We now examine the models' performance at the question level using four of the five metrics used to address RQ1 and RQ2—accuracy spread was excluded from this analysis since it only applies to the whole set of questions. Our goal is to identify the models' patterns of accuracy and stability over the 10 response for each question. Figure 2 shows four heatmap plots representing the models' per-question performance over the 10 responses for mean accuracy, pass@N, TAR@N, and cons@N metrics, respectively. In each plot, the six questions excluded from the ENADE score are highlighted in red.

In terms of mean accuracy (Fig. 2, top heatmap), Gemini 2.0 Flash consistently stands out as the most accurate model, achieving near-perfect or perfect scores on the majority of questions. Notably, it was the only model to answer Q09 correctly in all attempts, while all others failed. o1 also showed strong accuracy overall, but with more variability than Gemini 2.0 Flash, such as on Q11, Q18, and Q24. However, o1 achieved a perfect score (mean accuracy = 1.0) in some questions in which Gemini 2.0 Flash got only a few answers correct, like Q14, Q27, and Q34. DeepSeek-R1 maintained moderate accuracy, frequently surpassing GPT-4o but trailing Gemini 2.0 Flash and o1. DeepSeek-R1’s accuracy was particularly strong on Q26 and Q28, where it got all answers correct while both Gemini 2.0 Flash and o1 were less accurate. GPT-4o struggled with several questions, including Q09, Q11-13, Q16, Q19, and Q23-26, showing mean accuracy frequently below 0.3.

The pass@N per-question results (Fig. 2, second heatmap) reinforces the trends observed in the mean accuracy heatmap while also revealing divergence in robustness under repeated sampling. Gemini 2.0 Flash and o1 scored pass@N on nearly every question, demonstrating that even when not consistently correct, they often included the right answer at least once. DeepSeek-R1 also performs well, although slightly less consistently than Gemini 2.0 Flash and o1. GTP-4o, however, failed to generate a single correct answer for several questions, including Q16, Q19, Q23, and Q28, which were not excluded from the exam’s official score. This behavior is evidence of GPT-4o’s unreliability even on “non problematic” questions. In general, failed pass@N values, especially when they co-occur across models, such as on Q13, Q16 and Q21, indicate tricky questions that challenge the models even under repeated sampling.

TAR@N (Fig. 2, third heatmap) reveals that all models achieved full determinism in at least a few questions. Interestingly, Gemini 2.0 Flash and DeepSeek-R1 produced fully deterministic responses for the same number of questions (17 out of 25), more than the other models, even though DeepSeek-R1 was only the third best model in the accuracy metrics. This further supports the claim that consistency does not always correlate with correctness. GPT-4o was again the most inconsistent, achieving full determinism in only seven questions.

Finally, regarding cons@N (Fig. 2, bottom heatmap), Gemini 2.0 Flash again dominates, achieving full scores on 21 out of the 25 questions. o1 follows closely with 20 questions, while DeepSeek-R1 lags behind with only 17 questions. GPT-4o’s performance is notably low in this metric, reaching a consensus on a correct answer on only 10 questions. Across all models, questions Q13, Q21, and Q25 exhibit failed cons@N values, further signaling their potential ambiguity or misleading design. Conversely, Q10, Q14, Q15, Q18, Q22, Q30, Q34 and Q35 achieve cons@N = 1.0 across the board, suggesting their status as the easiest and most straightforward questions of the exam—at least from the four models’ perspective!

Curiously, all models failed to correctly answer Q16, a logic circuit interpretation question, which was not excluded from the official exam score. In fact, most models (with the exception of Gemini 2.0 Flash) answered this question

with $\text{TAR@N} = 0$, indicating weak internal consistency. Manual inspection suggests that visual parsing errors played a key role: the logical circuits are overlaid on top of multiple grid lines, causing the models to misidentify the gates or misunderstand their connections. This case illustrates how some visual layouts that are easily interpreted by humans can severely hinder multimodal reasoning, even for advanced models like Gemini 2.0 Flash and o1.

4.4 RQ4: Excluded Questions

To better understand how the four LLMs respond to questions deemed statistically problematic by ENADE organizers, we now focus on the six questions that were excluded from the official score: Q12, Q13, Q17, Q21, Q25, and Q32 [11]. Again, we base our analysis on the per-question results for the four metrics displayed in Fig. 2.

Q13 has the most striking results among the six excluded questions: all four models answered it incorrectly in every attempt, yielding zero mean accuracy and pass@N . Yet, the TAR@N score was 1.0 for every model, indicating that all models consistently gave the same wrong answer across all attempts. This suggests the presence of a strong distractor and highlights that high determinism (as measured by TAR@N) not only does not necessarily imply correctness, but can also coincide with systematic error.

In contrast, Q12 was answered correctly by most models, with three models (excluding GPT-4o) achieving top scores in pass@N and cons@N . Q17 and Q32 also showed similar results, with at least three models achieving pass@N and two models (Gemini 2.0 Flash and o1) achieving cons@N . These results suggest that any issues that may have caused these three questions to be excluded from the exam (e.g., semantic ambiguity or lack of clarity) did not pose any major challenge to the models that answered them correctly and consistently in our study.

Finally, Q21 and Q25 emerged as genuinely challenging across most models. For Q21, only one model (Gemini 2.0 Flash) reached top scores in pass@N and cons@N . Similarly, for Q25, only Gemini 2.0 Flash and o1 achieved pass@N , while only Gemini 2.0 Flash achieved cons@N . These results indicate that these two questions were generally difficult for most models, leading to overall low accuracy and consensus. This aligns with the psychometric analysis that led to their exclusion from the official score.

In summary, some questions, like Q10 and Q14, were reliably solved by all models, while others—such as Q09 and Q16—exposed significant weaknesses, including high response variability and failed consensus. On the six excluded questions, most models uniformly struggled on Q13, Q21, and Q25, aligning with their exclusion from the exam score. Other excluded questions such as Q12 and Q17 were handled well by some models, suggesting that psychometric exclusion does not always reflect model performance. Overall, Gemini 2.0 Flash was the most resilient model across the six excluded questions, achieving cons@N on five of them. GPT-4o, again, had the weakest performance, struggling across both

accuracy and stability dimensions, achieving pass@N on only two questions (Q17 and Q32) and failing to achieve cons@N in all of them.

5 Discussion

5.1 Implications for Stakeholders

Our study offers relevant takeaways for multiple stakeholders involved in the development, deployment, and evaluation of LLMs in educational contexts.

For LLM developers and researchers, our results highlight the importance of evaluating models under repeated sampling rather than relying on single responses. The observed differences between mean accuracy and pass@N or cons@N suggest that models may “know” the correct answer but fail to generate it reliably. Models like GPT-4o, for instance, showed highly inconsistent outputs that reduced trust in their ability to respond correctly even when they occasionally knew the answer. These findings emphasize the need for improved prompting strategies, robustness-focused training, and new decoding mechanisms that prioritize stability without sacrificing diversity [26].

For LLM users in educational contexts, our findings show important behavioral differences. Gemini 2.0 Flash and o1 were more accurate and stable, making them better candidates for high-stakes tasks. In contrast, GPT-4o’s erratic behavior can mislead users who rely on single outputs. Practitioners should be aware of this variability and use ensemble or majority-vote strategies to improve LLM reliability [17].

For exam designers, our work suggests LLM-based metrics can complement psychometrics in identifying flawed questions. The models’ collective failure on Q13 and confusion on Q21 and Q25 align with their official exclusion, supporting the use of automated diagnostics. Conversely, the models’ strong performance on excluded questions like Q12 and Q17 indicates that human statistical anomalies do not always reflect semantic problems. Thus, LLMs can serve as a second-opinion mechanism to flag questions for human review.

For educators and students, our findings provide direct evidence of the types of questions current LLMs handle well (e.g., Q10, Q15) versus poorly (e.g., Q13, Q16). Educators can use these insights to design better teaching strategies and prepare students for common assessment pitfalls.

Finally, our results show accuracy and stability are distinct dimensions. A model can be accurate but inconsistent (like o1) or consistent but wrong (like all models on Q13). This duality has major implications for LLM benchmarking, particularly in educational domains where consistency is often as crucial as correctness.

5.2 Limitations and Threats to Validity

While our study offers a detailed and multi-dimensional evaluation of four frontier LLMs on a high-stakes educational benchmark, it is not without limitations.

First, our evaluation used only the ENADE 2021 CS exam. Although this exam covers a comprehensive set of Computer Science questions, our findings may not generalize to other assessments without further validation.

Second, we used 10 generations per question per model to capture variability and measure stability, which is an arbitrary number. Larger sampling sizes (e.g., 50 or 100 generations) might yield more accurate estimates of model entropy and convergence but would also incur greater computational cost. Our choice of 10 strikes a balance between methodological rigor and practical feasibility, but future work could explore how sensitive our metrics are to this sampling depth.

Third, with the exception of Q16, we did not investigate the causes of the models' incorrect answers, including content misunderstandings, lack of visual acuity, ambiguity in question wording, or limitations in knowledge representation. A full qualitative analysis of the models' reasoning was beyond our scope due to space constraints, but we consider this an important direction for future work.

Fourth, we did not conduct statistical significance tests to verify if model differences were meaningful. Our goal was a comparative, multi-metric overview rather than making inferential claims, but formal statistical analysis is a key area for future work.

Fifth, our methodology does not control for model differences in architecture, training data, or fine-tuning procedures. Given that the ENADE 2021 exam was public well before the models' training cutoff dates, this raises the risk of data leakage. While some questions were consistently answered incorrectly, this does not eliminate the possibility of exposure, since LLMs can internalize content in ways that do not lead to verbatim memorization, and such exposure could bias comparative results. Without access to the models' training data, this confounding factor cannot be ruled out.

Sixth, our results rely on the assumption that the ENADE answer key is authoritative and unambiguous. Although this is reasonable for a national-level standardized exam, it is possible that some questions have more than one defensible interpretation or that the models' incorrect responses are partially justifiable [18]. Without access to item development documentation or human annotations of question clarity, we cannot verify the semantic validity of each item in detail.

Finally, we only used image-based inputs, which reflects the exam's original format but may limit models with weaker visual perception. Evaluating model performance on text-only or hybrid inputs is left for future work.

6 Conclusion and Future Work

Our evaluation of four frontier LLMs on the 2021 ENADE CS exam found that Gemini 2.0 Flash was the most accurate and stable, while GPT-4o had the lowest accuracy and greatest variability. We also found that questions excluded from the official score due to psychometric issues were often problematic for the models as well, reinforcing the value of using multi-run stability metrics as a complementary tool for exam validation.

We envision three interesting directions for future work. First, we will evaluate newer models (e.g., o3, Gemini 2.5 Pro) to determine if the accuracy and stability limitations we observed are mitigated. Second, we will analyze model performance on ENADE exams across other knowledge domains to check whether their behavior is discipline-sensitive. Finally, we will complement our quantitative results with a qualitative investigation into model reasoning and error causes, aiming to understand not just *what* the models get wrong, but *why*.

Acknowledgments. The generative AI tools ChatGPT, Gemini, and GitHub Copilot were used to assist in analyzing our experimental data and revising the manuscript. Nabor C. Mendonça is partially supported by Brazil’s National Council for Scientific and Technological Development (CNPq) under grant 313558/2023-0.

Artifact Availability. The data and code used in this study are publicly available at <https://anonymous.4open.science/r/LLMENADEBRACIS2025-62D5/>.

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